work, the bulk description and the boundary description are on an absolutely equal footing. One is not primary and the other secondary. In much the same spirit as the relation between the five string theories, the bulk and boundary theories are translations of each other. The unusual feature of this particular translation, though, is that the bulk theory has more dimensions than the equivalent theory formulated on the boundary. Moreover, whereas the bulk theory includes gravity (since Maldacena formulated it using string theory), calculations show that the boundary theory doesn't. Nevertheless, any question asked or calculation done in one of the theories can be translated into an equivalent question or calculation in the other. While someone unfamiliar with the dictionary would think that the corresponding questions and calculations have absolutely nothing to do with each other (for example, since the boundary theory does not include gravity, questions involving gravity in the bulk theory are translated into very-different-sounding, gravity-less questions in the boundary theory), someone well versed in both languages—an expert on both theories—would recognize their relationship and realize that the answers to corresponding questions and the results of corresponding calculations must agree. Indeed, every calculation done to date, and there have been many, supports this assertion.

The details of all this are challenging to grasp fully, but don't let that obscure the main point. Maldacena's result is amazing. He found a concrete, albeit hypothetical, realization of holography within string theory. He showed that a particular quantum theory without gravity is a translation of—is indistinguishable from—another quantum theory that includes gravity but is formulated with one more space dimension. Vigorous research programs are under way to determine how these insights might apply to a more realistic universe, our universe, but progress is slow as the analysis is fraught with technical hurdles. (Maldacena chose the particular hypothetical example he did because it proved relatively easy to analyze mathematically; more realistic examples are much harder to deal with.) Nevertheless, we now know that string theory, at least in certain contexts, has the capacity to support the concept of holography. And, as with the case of geometric translations described earlier, this provides yet another hint that spacetime is not fundamental. Not only can the size and shape of spacetime change in